National University of Computer and Emerging Sciences
Karachi Campus

Digital Logic Design (EE1005)

Date: 20th May, 2026

Course Instructors:

Dr. Rabia Tabassum, Sir Javed Quershi, Dr. M Adeel, Ms. Hira Tunio, Ms. Sana Naz, Mr. Ishtiaq Ahmed, Mr. M Miqdad Khan, Mr. M Rahim, Mr. Abuzar Zafar, and Dr. Muhammad Nouman Durrani

Final Exam

Total Time (Hrs): 3
Total Marks: 88
Total Questions: 5

Roll No Section Student Signature

Do not write below this line

Attempt all the questions in the given sequence.

CLO # 2: Demonstrate the acquired knowledge to apply techniques related to the design and analysis of digital electronics circuits, including Boolean Algebra and Multi-variable Karnaugh map methods.

Design and analysis of digital electronics circuits.
[4 X 4 = 16 Marks]

Time: 30 Minutes

Q1: (a) A digital system receives the following numbers in different bases: $(527)_8$,$(101101101)_2$,$(2F3)_{16}$. Convert all numbers to decimal and compute: $(527)_8+(101101101)_2-(2F3)_{16}$ **= $343_{10} + 365_{10} - 755_{10}$**

= -47_{10}

(b) Consider the following 8-bit signed binary numbers: A=10110110 B=01101101. Convert both binary numbers A and B into their signed decimal equivalents. Also, compute the operation A - B using 2's complement arithmetic, and state whether an overflow occurs.

A 's MSB is 1, so A is a **Negative Number**. A = -128 + 32 + 16 + 4 + 2 = -74

B's MSB is **0**, so B is a **Positive Number**. B= 64 + 32 + 8 + 4 + 1 = 109

To compute A - B using 2's complement arithmetic, we rewrite the subtraction as an addition problem:

A - B = A + (-B) -B = 10010011$.

```
Carry:  1 1 1 1  1 1
   A:  1 0 1 1  0 1 1 0  ( -74 )
 + -B:  1 0 0 1  0 0 1 1  (-109 )
 ------------------------
 Total:1 0 1 0 0  1 0 0 1
```

Discarding the extra 9th bit (the final carry-out of 1), our 8-bit result is: Result = **01001001**

(c) Simplify completely using Boolean identities and DeMorgan's theorem:

$$F(A,B,C)=[(A+B')(A'+C)]'+AB+AC$$

Show all intermediate steps. State the Boolean law used at each step. Also, implement the simplified Boolean function using NOR gates only.

$$F(A,B,C) = [(A+B')(A'+C)]' + AB + AC$$

Step-by-Step Reduction

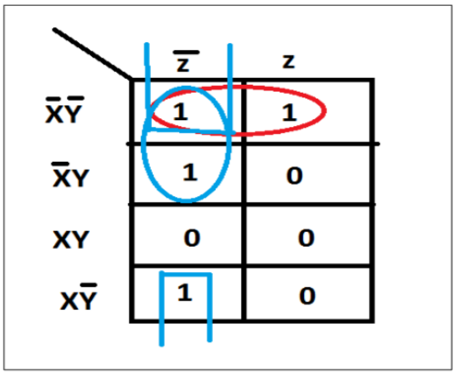
Step	Expression	Boolean Law / Identity Used
Given	$F = [(A+B')(A'+C)]' + AB + AC$	
1	$F = (A+B')' + (A'+C)' + AB + AC$	De Morgan's Theorem ($[X \cdot Y]' = X' + Y'$)
2	$F = (A' \cdot B'') + (A'' \cdot C') + AB + AC$	De Morgan's Theorem ($[X+Y]' = X' \cdot Y'$)
3	$F = A'B + AC' + AB + AC$	Double Involution Law ($X'' = X$)
4	$F = (A'B + AB) + (AC' + AC)$	Commutative / Associative Law (Rearranging terms)
5	$F = B(A'+A) + A(C'+C)$	Distributive Law
6	$F = B(1) + A(1)$	Complement Law ($X + X' = 1$)
7	$F = B + A$	Identity Law ($X \cdot 1 = X$)
8	$F = A + B$	Commutative Law

Draw NOR Gate of it

(d) In a water pumping station there are three pumps (x, y, z) that are to be monitored. If x=1 it implies that pump x is on and working. You are required to design a combinational circuit to raise an alarm when the majority of the pump fails. Derive the minimal Boolean expression for this combinational circuit using K-map and construct the logical circuit diagram.

Truth Table

X	Y	Z	Alarm
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0



K-map

Equation: X'Y' + X'Z' + Y'Z'

```
    yz
x   00    01    11    10
  +-----+-----+-----+-----
0 |  1  |  1  |  0  |  1
  +-----+-----+-----+-----
1 |  1  |  0  |  0  |  0
```

$$A = \bar{x}\bar{y} + \bar{y}\bar{z} + \bar{x}\bar{z}$$

CLO # 3 Analyze small –scale combinational digital circuits.

Design and analysis of Combinational Logic Circuits: **[4 X 4 = 16 Marks]**

Time: 30 Minutes

Q2: (a) Design a 4-bit binary adder–subtractor circuit using only full adders and a single mode control input M. The circuit should perform binary addition when M=0 and binary subtraction using 2's complement when M=1. Also, draw the complete logic diagram of the circuit.

```
     ┌──────────┐              ┌──────────┐              ┌──────────┐              ┌──────────┐
B0 ─┤          ├─ B0'    B1 ─┤          ├─ B1'    B2 ─┤          ├─ B2'    B3 ─┤          ├─ B3'
M ──┤   XOR    │         M ──┤   XOR    │         M ──┤   XOR    │         M ──┤   XOR    │
     └──────────┘              └──────────┘              └──────────┘              └──────────┘
```

Date.

Addition Module:

$M = 0$

$B_i \oplus 0 = B_i$

$C_{in} = 0$

$A + B$ is performed

Subtraction Module:

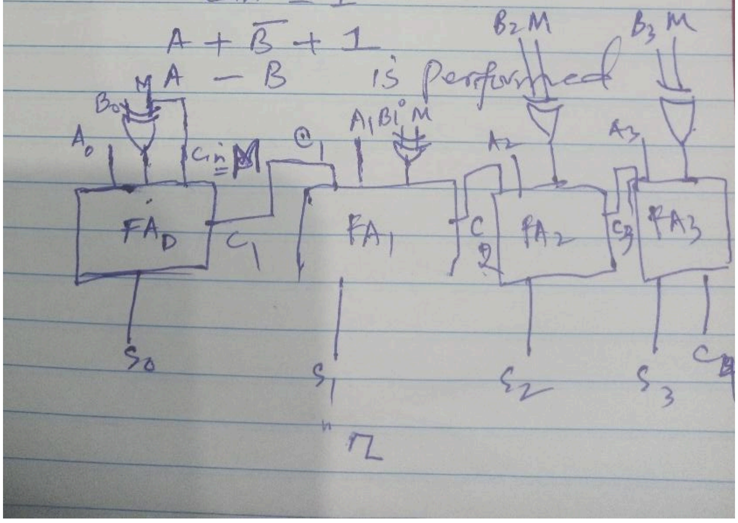
$M = 1$

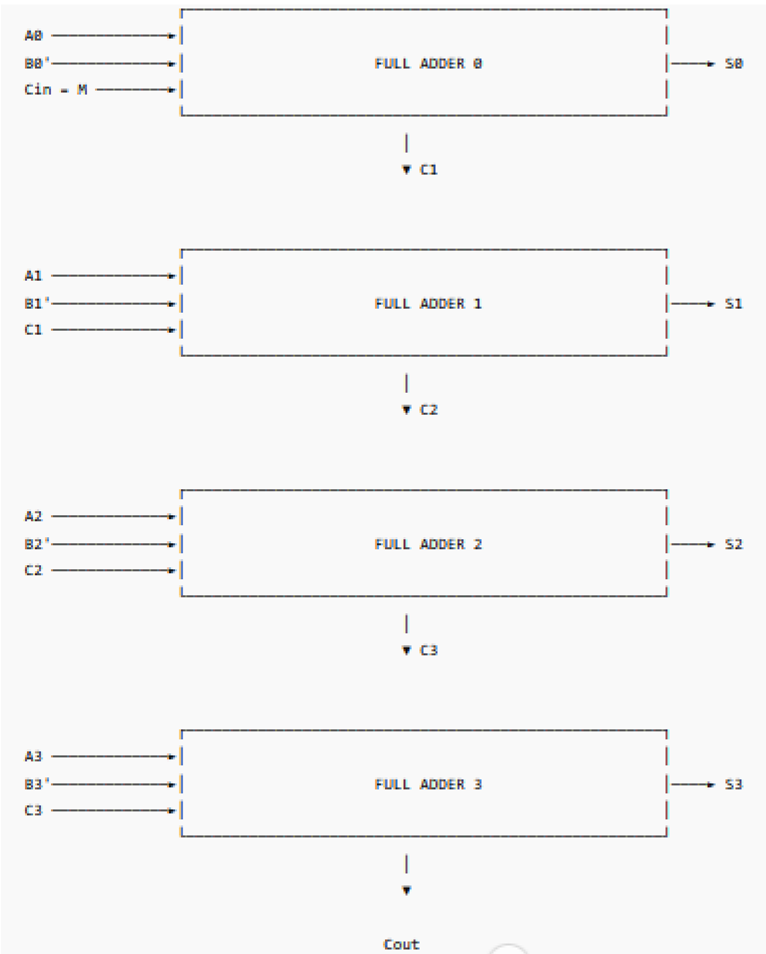
$B_i \oplus 1 = \overline{B_i}$

$C_{in} = 1$

$A + \overline{B} + 1$

$A - B$ is performed





(b) A 3-to-8 decoder has one faulty output line (consider line 5) permanently stuck at logic 0. Design an alternative logic arrangement using additional gates so that all minterms can still be generated correctly.

- **Lines 0–4 and 6–7:** Retained directly from the original decoder outputs.
- **Line 5:** Disconnected from the faulty pin 5, and replaced by the output of the new external **AND gate** (A2A1’A0).

(c) Design a circuit using **4×1 multiplexers** that performs the following operations based on control inputs S_1,S_0:

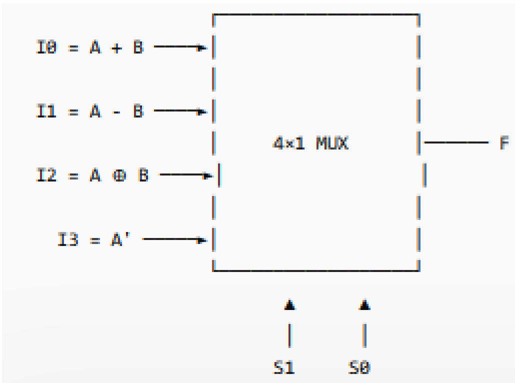
S_1S_0	Operation
00	A+B
01	A−B
10	A⊕B
11	A'

$$I_0 = A + B$$

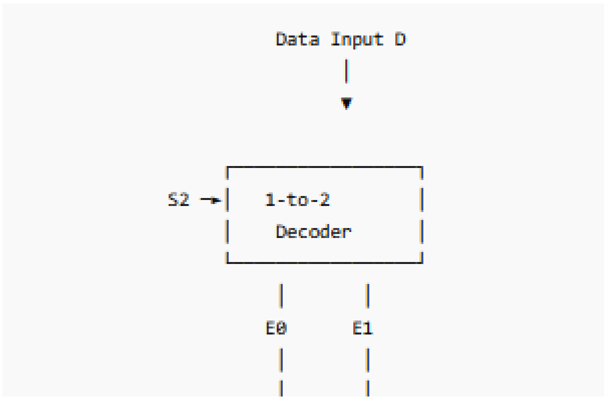
$$I_1 = A - B$$

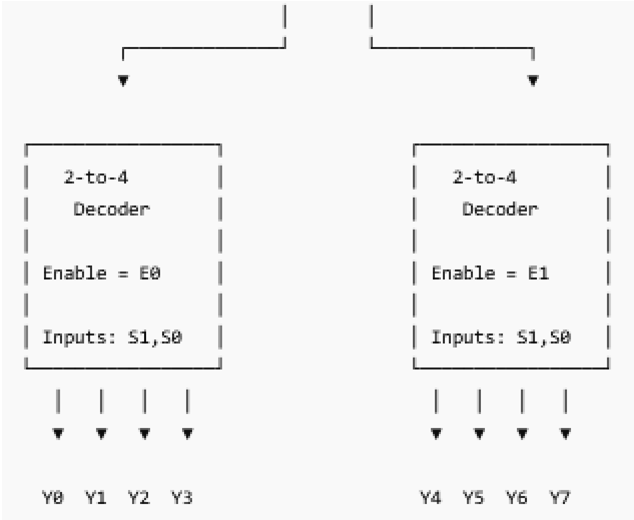
$$I_2 = A \oplus B$$

$$I_3 = A'$$



(d) Show how to implement a 1-8 demultiplexer using the minimum number of 2-4 and 1-2 decoders with enable inputs and nothing else.





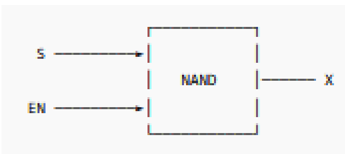
CLO # 4 Design small-scale combinational and synchronous sequential digital circuit using Boolean Algebra and K-map.

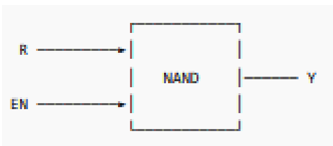
Design and analysis of Sequential Logic Circuits: **[4 X 4 = 16 Marks]**

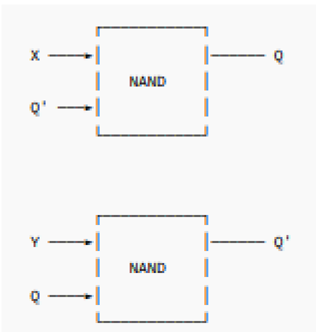
Time: 40 Minutes

Q3: (a) Draw the complete gate-level schematic of a Gated SR Latch using only 2-input NAND gates. Clearly label the inputs (S, R, EN) and the outputs (Q, Q'). Assume the Gated SR Latch designed above is initially in a stable Reset state (Q = 0, Q' = 1). Owing to a hazard caused by the positive-edge clock signal driving the circuit, the input signals experience a sequence of rapid successive transitions. The events occur in the exact chronological order detailed below:

t_1: S switches from 0 to 1. (R is 0, EN is 0). **t_2:** EN switches from 0 to 1.
t_3: R switches from 0 to 1. (Now S=1, R=1, EN=1). **t_4:** EN switches from 1 to 0.
t_5: S and R both simultaneously switch from 1 to 0.
For each time interval (t_1 □ t_2, t_2 □ t_3, t_3 □ t_4, and t_4 □ t_5), determine the logical values of the internal outputs of the enabling NAND gates, as well as the final outputs Q and Q'.







Where:

- $X = (S \cdot EN)'$
- $Y = (R \cdot EN)'$

Initial State:

$$Q = 0, \quad Q' = 1$$

Interval	X	Y	Q	Q'
Initial	1	1	0	1
$t_1 \to t_2$	1	1	0	1
$t_2 \to t_3$	0	1	1	0
$t_3 \to t_4$	0	0	1	1
$t_4 \to t_5$	1	1	?	?
After t_5	1	1	?	?

Q = 0 0 1 1 ? ?

Q'= 1 1 0 1 ? ?

(b) Analyze the following timing diagram for a Gated D Latch. The latch is initially storing a value of Q = 1 at t_0. Reconstruct and accurately draw the waveform for the output Q from t_0 to t_9. Does the glitch in D right at t_0 affect the output Q? Why or why not? [Please solve here on the question paper.]

```
           t0   t1   t2   t3   t4   t5   t6   t7   t8   t9
           |    |    |    |    |    |    |    |    |    |
EN   _____/‾‾‾‾‾‾‾‾‾\____/‾‾‾‾‾‾‾‾‾\____/‾‾‾‾‾\__________
           |    |    |    |    |    |    |    |    |    |
D    __/\_/‾‾‾‾\____/‾‾‾‾\____/‾‾‾‾\___________/‾‾‾‾‾‾\___
       (glitch)
           |    |    |    |    |    |    |    |    |    |
```

1 0 0 0 1 1 0 0 0

No effect as Enable is Low. Here we are considering if EN is 1 then D pattern will be considered.

(c) Two edge-triggered J-K flip-flops are shown in Fig. 2 below. If the inputs are as shown, draw the Q output of each flip-flop relative to the clock, and explain the difference between the two. The flip-flops are initially RESET. [Please solve here on the question paper.]

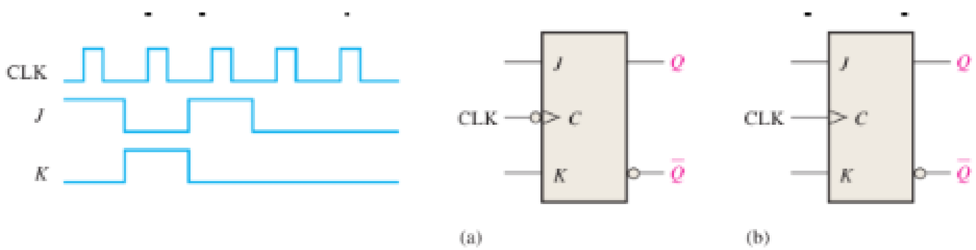
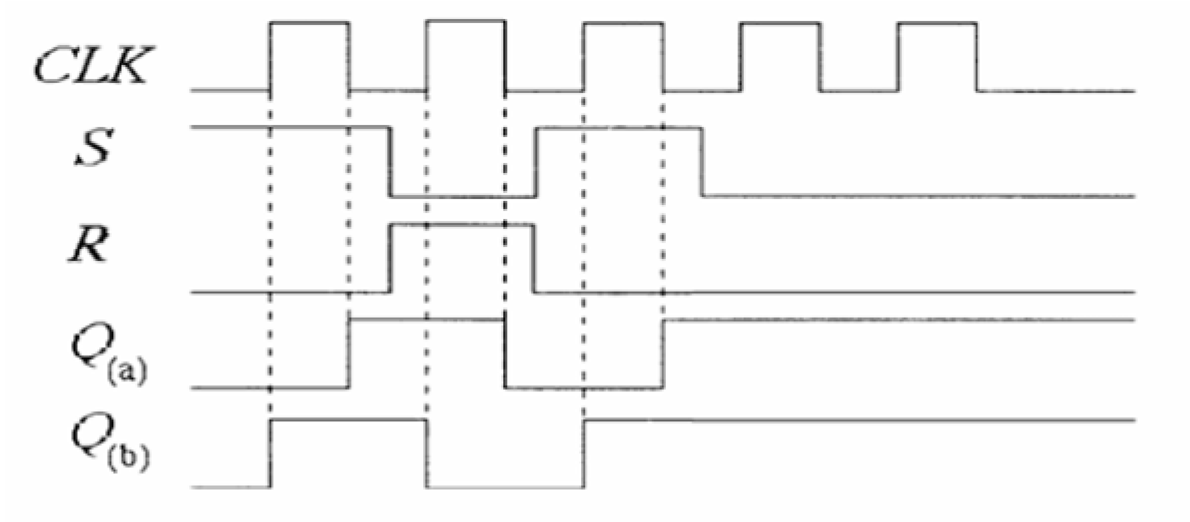


Fig. 2



(d) For the circuit in Fig. 3, develop a timing diagram for eight clock pulses, showing the QA and QB outputs in relation to the clock

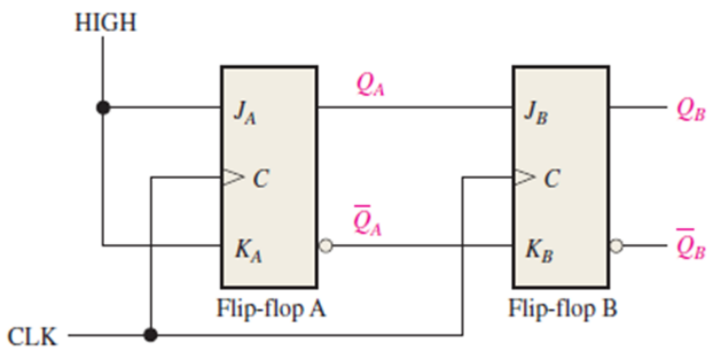


Fig. 3.

From the figure:

Flip-Flop A

- $J_A = 1$
- $K_A = 1$

Therefore Flip-Flop A toggles on every clock pulse.

Hence:

$$Q_A = 01010101\ldots$$

Inputs:

- $J_B = Q_A$
- $K_B = \overline{Q_A}$

For JK FF:

- If $J = 1, K = 0 \rightarrow$ SET
- If $J = 0, K = 1 \rightarrow$ RESET

Thus:

- When $Q_A = 1$, Q_B becomes 1
- When $Q_A = 0$, Q_B becomes 0

Hence:

$$Q_B = Q_A$$

Assume initially:

$$Q_A = 0, \quad Q_B = 0$$

Clock Pulse	(QA)	(QB)
Initial	0	0
1	1	1
2	0	0
3	1	1
4	0	0
5	1	1
6	0	0
7	1	1
8	0	0

```
CLK : _|‾|_|‾|_|‾|_|‾|_|‾|_|‾|_|‾|_|‾|_|‾|_

QA  : __‾‾__‾‾__‾‾__‾‾__‾‾__‾‾__‾‾__‾‾__

QB  : __‾‾__‾‾__‾‾__‾‾__‾‾__‾‾__‾‾__‾‾__
```

CLO # 4 Design small-scale combinational and synchronous sequential digital circuit using Boolean Algebra and K-map.

Design and analysis of Sequential Logic Circuits: [6 + 4 + 4 + 8 = 22 Marks]

Time: 40 Minutes

Q4: (a) Design a counter with the irregular binary count sequence shown in the state diagram of Fig: 4. Use D flip-flops.

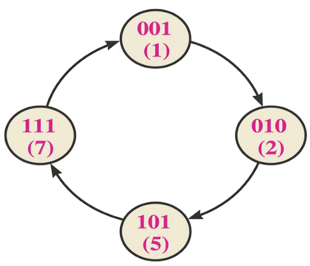
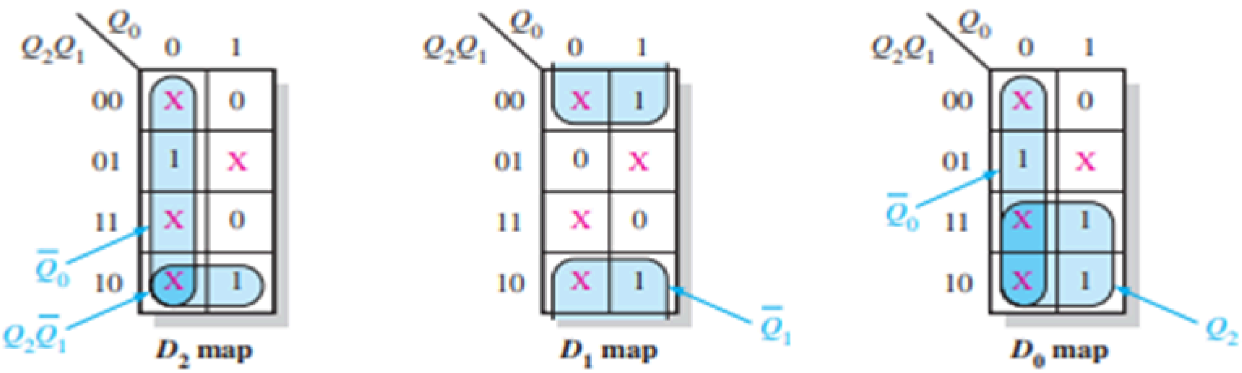


Fig. 4

Next-state table.

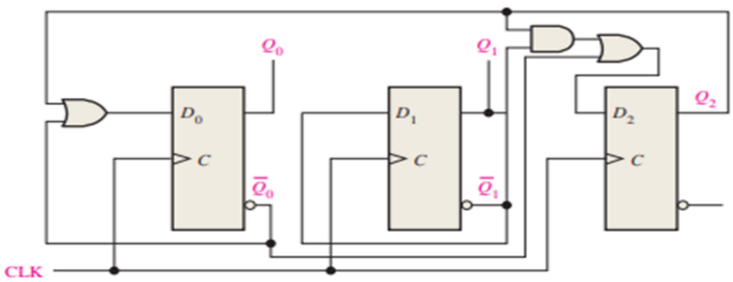
Present State			Next State		
Q_2	Q_1	Q_0	Q_2	Q_1	Q_0
0	0	1	0	1	0
0	1	0	1	0	1
1	0	1	1	1	1
1	1	1	0	0	1

The D inputs are plotted on the present-state Karnaugh maps in Fig: 2. Also “don’t cares” can be placed in the cells corresponding to the invalid states of 000, 011, 100, and 110, as indicated by the red Xs



Group the 1s, taking advantage of as many of the “don’t care” states as possible for maximum simplification, as shown in Fig: 2 . The expression for each D input taken from the maps is as follows:

$$D_0 = \overline{Q}_0 + Q_2$$
$$D_1 = \overline{Q}_1$$
$$D_2 = \overline{Q}_0 + Q_2\overline{Q}_1$$



(b) If a 5-bit ring counter has an initial state 10101, determine the waveform for each Q output.

Clock Pulse	($Q_4Q_3Q_2Q_1Q_0$)
Initial	10101
1	11010
2	01101
3	10110
4	01011
5	10101 (repeats)

Suppose for 6 clock cycles. The Students must go for at least 3-4 cycles

Clock : ─┐ ┐ ┐ ┐ ┐ ┐

Q0 : 1 0 1 0 1 1
Q1 : 0 1 0 1 1 0
Q2 : 1 0 1 1 0 1
Q3 : 0 1 1 0 1 0
Q4 : 1 1 0 1 0 1

(c) A 4-bit synchronous counter uses JK flip-flops with:

- ► Flip-flop delay = 10 ns ► Gate delay = 5 ns

Determine the maximum operating frequency of the counter.

F= 1/ t_{ff} + t_{gd} = 1/10ns+5ns = 1/15 x 10^{-9} = 66.7 x 10^{6} HZ= 66.7 MHz

(d) Design a synchronous 3-bit binary up/down counter that follows the non-sequential pattern. The direction of the count is controlled by an input variable, M:

- When **M = 1 (Up)**, the counter follows the sequence: 0 □ 1 □ 3 □ 7 □ 6 □ 4 □ 2 □ 0.
- When **M = 0 (Down)**, the counter reverses the exact sequence: 0 □ 2 □ 4 □ 6 □ 7 □ 3 □ 1 □ 0.

Create a complete excitation table for the counter using three **JK Flip-Flops** (Q_2, Q_1, Q_0) and the control input M. Include the unused state (5) as a Don't Care condition (X).

Use Karnaugh Maps (K-maps) to derive the minimized sum-of-products (SOP) expressions for the flip-flop inputs. Draw the complete logic gate schematic for the synchronous counter.

M	Present State	Next State	J2	K2	J1	K1	J0	K0
1	000	001	0	X	0	X	1	X
1	001	011	0	X	1	X	X	0
1	011	111	1	X	X	0	X	0
1	111	110	X	0	X	0	X	1
1	110	100	X	0	X	1	0	X
1	100	010	X	1	1	X	0	X
1	010	000	0	X	X	1	0	X
1	101	XXX	X	X	X	X	X	X

M	Present State	Next State	J2	K2	J1	K1	J0	K0
0	000	010	0	X	1	X	0	X
0	010	100	1	X	X	1	0	X

M	Present State	Next State	J2	K2	J1	K1	J0	K0
0	100	110	X	0	1	X	0	X
0	110	111	X	0	X	0	1	X
0	111	011	X	1	X	0	X	0
0	011	001	0	X	X	1	X	0
0	001	000	0	X	0	X	X	1
0	101	XXX	X	X	X	X	X	X

From K-Map we get the following expressions. We can get any valid expression, based on students' approach towards picking cells from the k-map:

$$J_2 = Q_1Q_0$$

$$K_2 = Q_2Q_0'M + Q_2Q_1M'$$

$$J_1 = Q_2 + MQ_0$$

$$K_1 = Q_1(Q_2'M + Q_0M')$$

$$J_0 = MQ_1'Q_2' + M'Q_1Q_2$$

$$K_0 = MQ_1Q_2 + M'Q_1'Q_2'$$

CLO # 4 Design small-scale combinational and synchronous sequential digital circuit using Boolean Algebra and K-map.

Design and analysis of Sequential Logic Circuits: **[4 + 4 + 4 + 6 = 18 Marks]**

Time: 40 Minutes

Q5: (a) Consider the binary count sequence counter shown in Fig. 4. Assume that, due to noise, the counter may occasionally enter an invalid state. Design a self-correcting mechanism such that the counter automatically returns to the valid counting sequence, assuming the initial valid state is 001.

Valid sequence:

$$000 \to 001 \to 010 \to 011$$

Invalid states:

$$100, 101, 110, 111$$

Solution

Design the next-state logic such that:

$$100 \to 000$$

$$101 \to 000$$

$$110 \to 000$$

$$111 \to 000$$

instead of using don't cares.

If the student has solved until here, assign 2 marks, and Next we just to see the Next state table and K-map

(b) Design a 4-bit right-shift register. Now suppose that the 4-bit right-shift register initially contains: 1010. The serial input sequence is: 1, 1, 0, 1, 0. Determine register contents after each clock pulse.

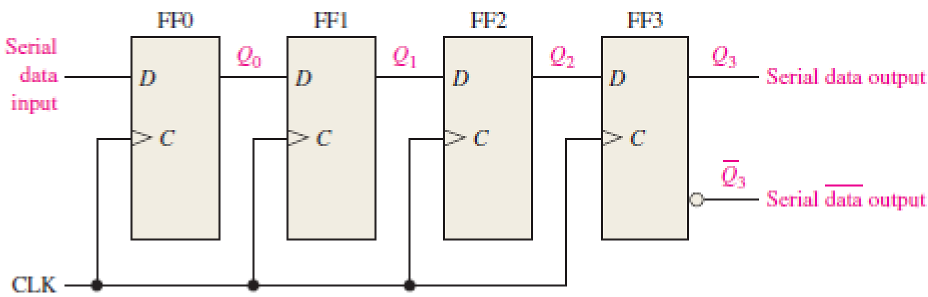


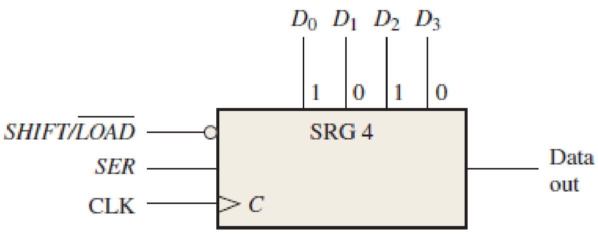
FIGURE 8–3 Serial in/serial out shift register.

Clock Pulse	Serial Input	Register Contents
Initial	—	1010
1	0	0101
2	1	1010
3	0	0101
4	1	1010
5	1	1101

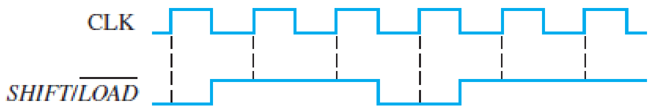
OR

Clock Pulse	**Serial Input**	**Register Contents**
Initial	—	1010
1	1	1101
2	1	1110
3	0	0111
4	1	1011
5	0	0101

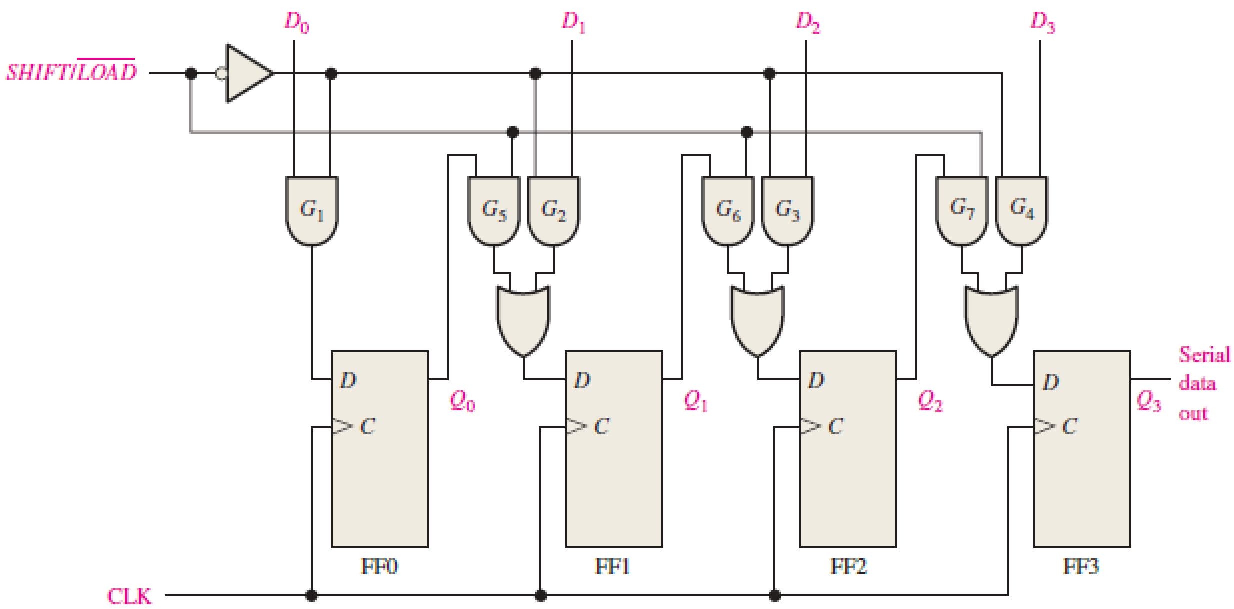
(c) Design a 4-bit parallel in/serial out shift register. Now suppose the shift register in Figure 5 (a) has SHIFT/$\overline{\text{LOAD}}$ and CLK inputs as shown in part (b). The serial data input (SER) is a 0. The parallel data inputs are D0 = 1, D1 = 0, D2 = 1, and D3 = 0 as shown. Develop the data-output waveform in relation to the inputs. [Please solve here on the question paper.]



(a)



(b)



(a) Logic diagram

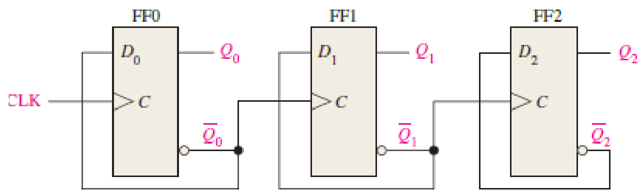
When *SHIFT*/***LOAD’*** is LOW, gates G_1 through G_4 are enabled, allowing each data bit to be applied to the *D* input of its respective flip-flop.
When *SHIFT*/***LOAD’*** is HIGH, gates G_1 through G_4 are disabled and gates G_5 through G_7 are enabled, allowing the data bits to shift right from one stage to the next.

- **Initial State (Before Clock Pulse 1):** The internal stages hold unknown or clear values.
- **Clock Pulse 1:** * Look at $SHIFT/\overline{LOAD}$: It is **LOW** (0) at this active clock edge.
  - **Action:** A parallel load occurs. The inputs ($D_3D_2D_1D_0 = 0101$) are loaded directly into the register stages ($Q_3Q_2Q_1Q_0$).
  - **Output Value:** The output pin (Q_3) immediately expresses the value of D_3, which is 0.
- **Clock Pulse 2:** * Look at $SHIFT/\overline{LOAD}$: It has transitioned to **HIGH** (1).
  - **Action:** The register performs a shift. The value from Q_2 moves to Q_3, Q_1 moves to Q_2, Q_0 moves to Q_1, and SER enters Q_0.
  - **Output Value:** Q_3 takes the old value of Q_2 ($D_2 = 1$). The output goes **HIGH** (1).
- **Clock Pulse 3:** * Look at $SHIFT/\overline{LOAD}$: It is **HIGH** (1).
  - **Action:** Another shift occurs.
  - **Output Value:** Q_3 takes the old value of Q_1 ($D_1 = 0$). The output drops **LOW** (0).
- **Clock Pulse 4:** * Look at $SHIFT/\overline{LOAD}$: It has dropped back to **LOW** (0).
  - **Action:** A parallel reload occurs. The inputs (0101) are reloaded into the register.
  - **Output Value:** Q_3 instantly displays the reloaded D_3 value, which is 0.
- **Clock Pulse 5:** * Look at $SHIFT/\overline{LOAD}$: It has gone back to **HIGH** (1).
  - **Action:** A shift operation occurs.
  - **Output Value:** Q_3 receives the value from Q_2 ($D_2 = 1$). The output goes **HIGH** (1).
- **Clock Pulse 6:** * Look at $SHIFT/\overline{LOAD}$: It remains **HIGH** (1).
  - **Action:** A shift operation occurs.
  - **Output Value:** Q_3 receives the value from Q_1 ($D_1 = 0$). The output drops **LOW** (0).

Summarized Final Solution:
Clock 1 2 3 4 5 6
0 □ 1 □ 0 □0 □ 1□0

(d) Design a 3-bit asynchronous up-counter using positive-edge-triggered D flip-flops.

(i) Draw the complete logic circuit diagram.



(ii) Modify your design from part (i) by adding the necessary decoding logic to detect and output a HIGH signal (1) specifically when the counter hits the numbers 3 and 5.

$$Y = (\overline{Q}_2 \cdot Q_1 \cdot Q_0) + (Q_2 \cdot \overline{Q}_1 \cdot Q_0)$$

(iii) Also, draw the original **4-bit** counter design to convert it into a Mod-6 asynchronous counter that resets to 000 after reaching its maximum valid state. Explain your logic and draw the modified reset circuitry.

A 3-bit counter is also acceptable here, as it would technically be sufficient.

- We must detect this state 0110 (or 110) immediately and use it to trigger the asynchronous Clear CLR inputs of the flip-flops, forcing the counter back to 0000 (or 000).
- Connect the outputs Q2 and Q1 to the inputs of a 2-input NAND gate, and then connect the output of this NAND gate to the active-low **Clear CLR** inputs of all four/three flip-flops.